

Computational Economics

Homework: Setting Up Git and Python

Create a repository, run a script and notebook, and verify the results

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Prepare Your Project

- ▶ Use the lab slides on [Introduction to Git](#) and [Introduction to Python](#).
- ▶ Download the [starter files](#): `markov_processes.py`, `markov_processes.ipynb`, `requirements.txt`, and `.gitignore`.
- ▶ Create a project folder named `computational-economics` and open it in VS Code.
- ▶ You may use graphical interfaces. Be able to explain what you changed and how you checked the result.

Git: Create and Connect Your Repository

1. Install Git. Confirm that VS Code detects it, or check with `git --version`.
2. Register for GitHub and complete account verification.
3. Initialize Git in your project folder. Add a README and the supplied `.gitignore`.
4. Inspect and stage the intended files, then make your first commit.
5. Use **Publish to GitHub** in VS Code to create and connect a remote repository. Choose its visibility and review the files before publishing.
6. Make a second meaningful change, commit, and push. Open the GitHub URL and confirm that both commits appear.

Python: Run a Script and Notebook

1. Install Python 3 and VS Code with Microsoft's Python and Jupyter extensions.
2. Copy the starter script and notebook into your project folder.
3. Use **Python: Create Environment**, select its interpreter, install the packages in `requirements.txt`, and run `markov_processes.py`.
4. Open `markov_processes.ipynb` and select the same environment as its kernel. Install `ipykernel` there if prompted.
5. Restart the kernel and run all cells. Check that both files give the next-month probabilities **[0.50, 0.50]**.

You can also practice with the [notebook in Colab](#). Complete the local setup and compare the results.

Change an Input and Check the Result

1. Change the job-finding probability from 0.20 to 0.40, keeping the other inputs fixed.
2. Rebuild the transition matrix and verify next-month probabilities **[0.38, 0.62]**.
3. Restart the notebook kernel and run all cells to check reproducibility.
4. Update your README with the Python version, run instructions, both results, and an explanation of the change.
5. Commit and push your work. Check the latest version on GitHub.

Completion Checklist

- ▶ Your GitHub repository URL and two visible commit identifiers or messages.
- ▶ A working `.py` script and `.ipynb` notebook with matching results.
- ▶ A README explaining setup, execution, results, and one check you performed yourself.
- ▶ Local environments, caches, credentials, and private data excluded from the repository.

Keep the repository link and completion evidence ready for the announced submission channel and deadline.